

Applied Data Science Course

Machine Learning

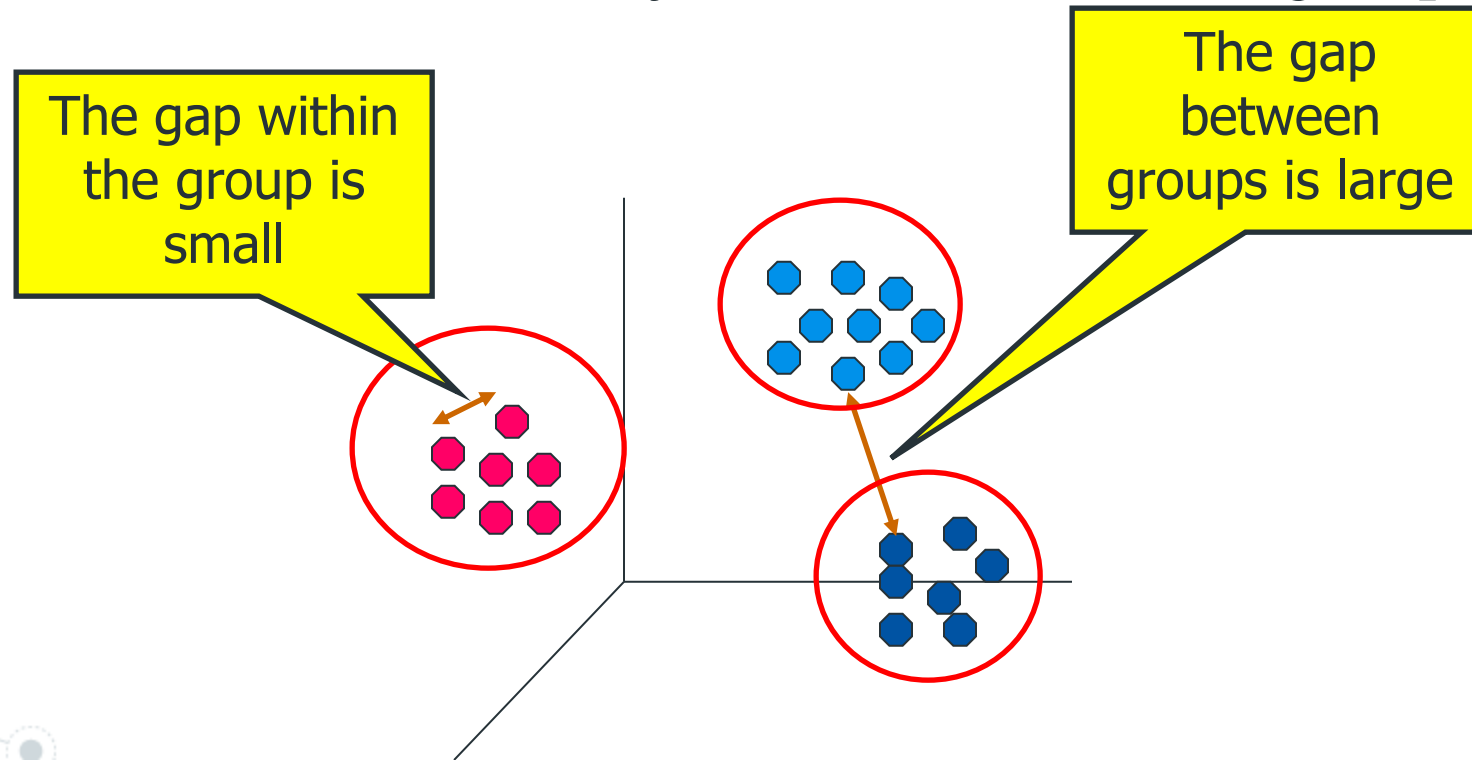
Unsupervised Learning

Le Ngoc Thanh
lnthanh@fit.hcmus.edu.vn
Department of Computer Science

Clustering

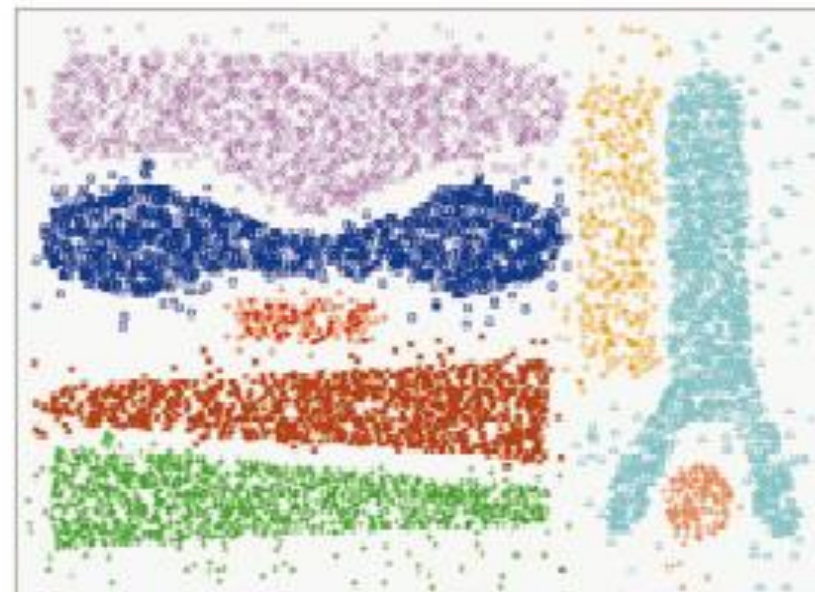
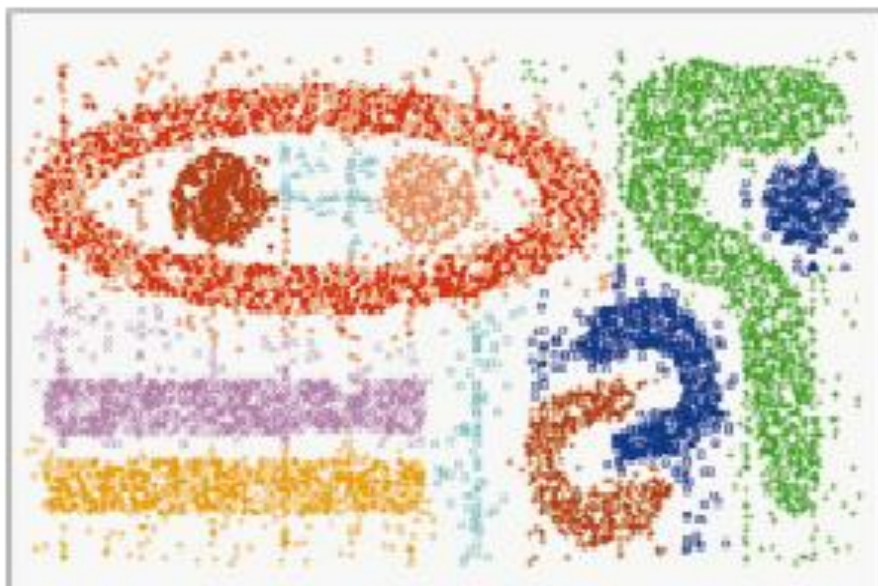
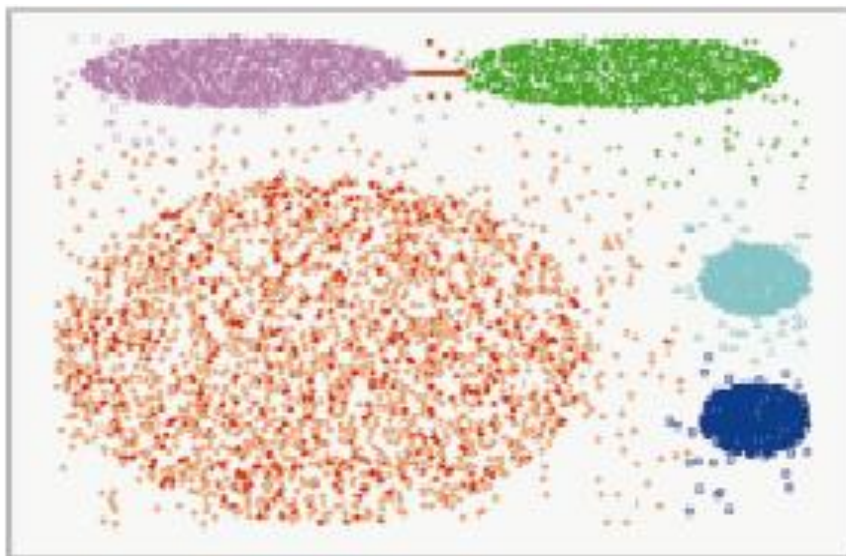
◎ Clustering is the process of grouping objects into clusters:

- The same group objects are highly similar.
- And very different from the subject in the rest of the groups.



Clustering

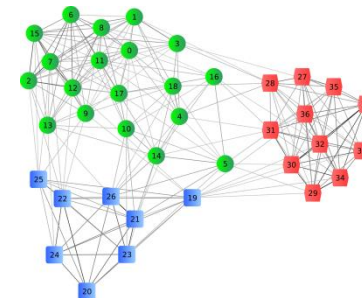
- ◎ Clustering is an unsupervised learning because the label/class is not pre-defined
- ◎ Therefore, clustering is a type of learning by visual rather than learning by examples



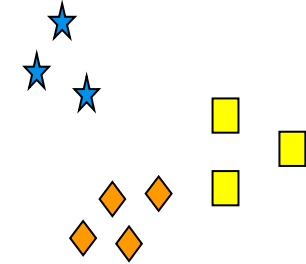
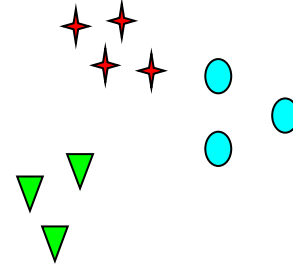
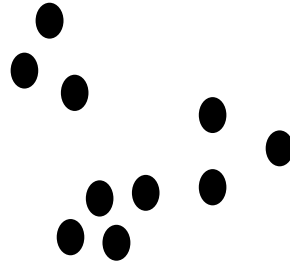
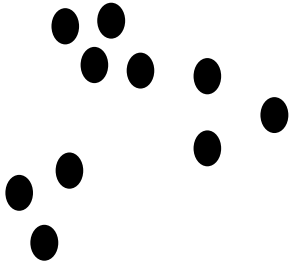
CHAMELEON

Some applications of clustering

- ◎ Grouping of related documents for web browsing
- ◎ Groups of genes and proteins have the same function
- ◎ Group of stocks with the same volatility
- ◎ Groups of areas of the same land type in geography
- ◎ Identify homegroups by Home type, value, and geographic location
- ◎ Defining a Gaming object group

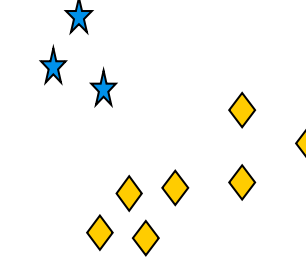
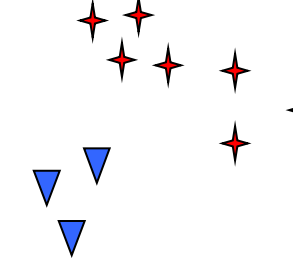
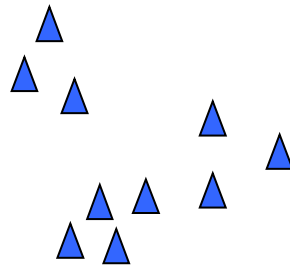
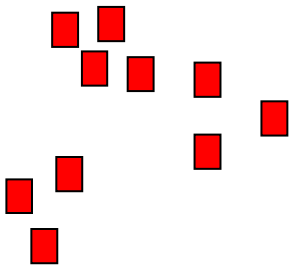


What is a group?



How many groups are there?

6 groups



2 groups

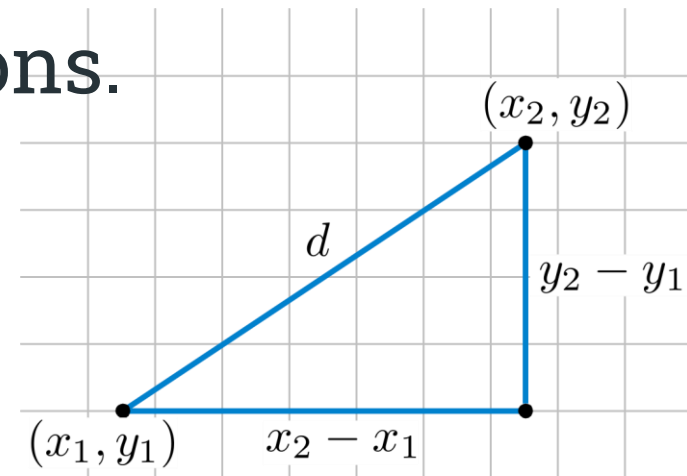
4 groups

A good Clustering?

- ◎ A good clustering method will have to create groups of high quality:
 - The same level in the High group.
 - Similar levels to other low-level groups.
- ◎ The quality of the clustering depends on:
 - Analog measurement
 - Its enforcement
 - Ability to discover some or all of the underlying patterns

Measure Similarity

- ◎ The distance used is mainly to measure the same or not similar level between the two objects.
 - Examples: Euclidean gap, Cosine Gap, Minkowski, Manhattan...
- ◎ Distance functions differ in value ranges, types, ranks, and variable elements.
- ◎ The weights of the dependent variables on the application and the data implications.



Some clustering methods

◎ Partitional clustering

- Formation of partitions and evaluate them based on some criteria
- Algorithms: K-Means, K-Medoids, CLARANS

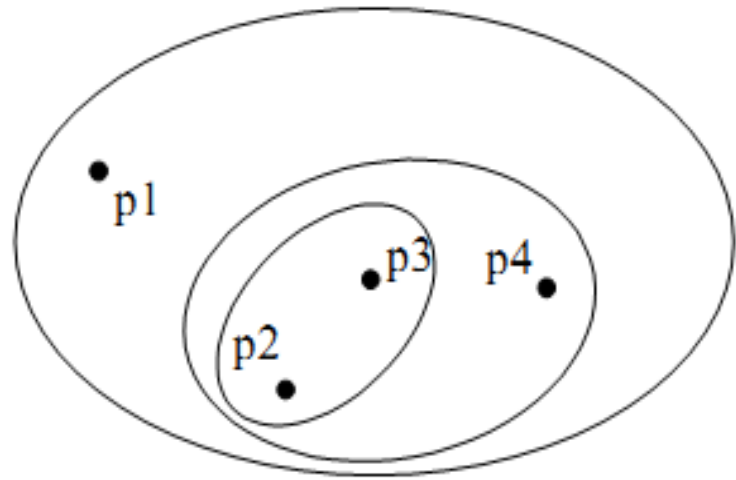
◎ Hierarchical clustering

- Create the division of Layers
Algorithms: Diana, Agnes, BIRCH, CAMELEON

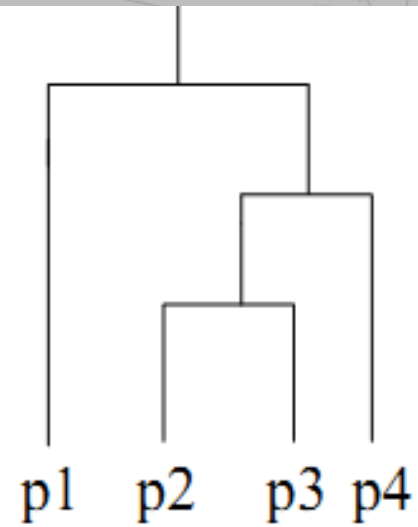
◎ Density-based clustering

- Based on the jaw and density function
Algorithms: DBSACN, OPTICS, DenClue

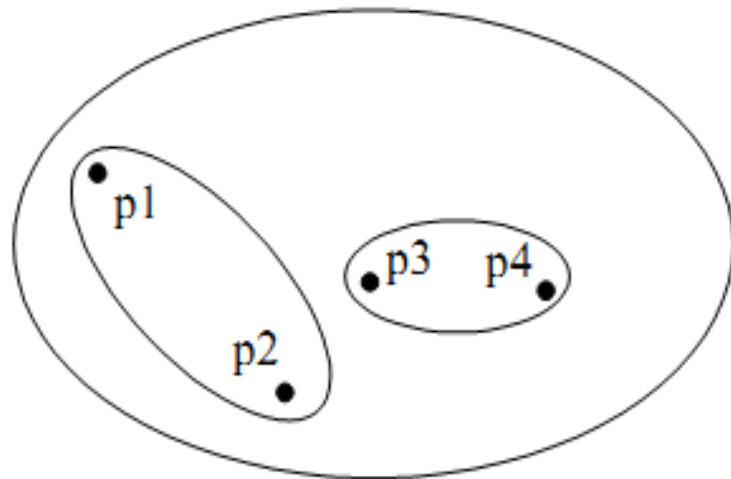
Example



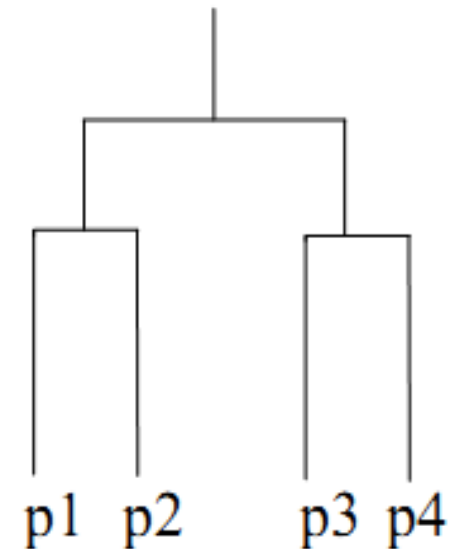
Traditional Hierarchical Clustering



Traditional Dendrogram



Non-traditional Hierarchical Clustering



Non-traditional Dendrogram

Some clustering methods (2/2)

◎ Grid-based methods

- Based on multi-level particle structure
Algorithms: STING, WaveCluster, CLIQUE

◎ Model-based methodology

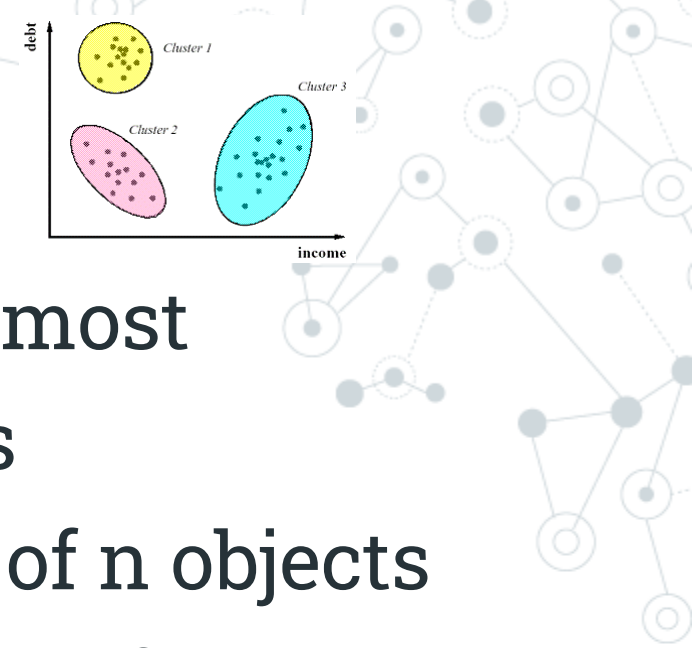
◎ Frequent pattern-based methods

◎ Methods based on binding or user guidance

◎ Link-based method

...

Partitioning Clustering



- ◎ Partitioning Clustering Is the simplest and most foundational method of clustering methods
- ◎ Idea: The partition of a D database consists of n objects into k Groups so that it optimizes the criteria of the partition.
- ◎ Global optimization: Full expression of all groups
- ◎ Heuristic algorithm:
 - K-means: each group is performed by the center value of the group
 - K-Medoids, PAM: each group is performed by one of the groups ' objects

K-Means algorithm (1/2)

- ◎ Before number k, each group is performed by the group's centroid value.
 - S1: Select Random K objects as the center of the groups
 - S2: Assigns each remaining object to the closest group based on distance measurement such as Euclidean, Cosine analogue, correlation,...
 - S3: Calculating the center value of each group based on newly joined objects.
 - S4: If the group Center has nothing to change or there are only a few points that change the group, stop, back to S2.



The End